

Sami Hatoum

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Education

- University College London** | *MSc Artificial Intelligence and Imaging* 2025 – 2026
Predicted Distinction
Dissertation: *Fair Uncertainty Quantification for Selective Prediction in Medical Image Segmentation*
Modules: Applied Deep Learning, Machine Learning in Medical Imaging, Statistical Modelling, Inverse Problems
- University of St Andrews** | *BSc Computer Science* 2021 – 2025
First Class Honours, Dean's List, Honours Project Distinction
Dissertation: *Sokoban Procedural Instance Generation* (constraint optimisation over a PSPACE-complete design space)
Modules: Computational Complexity, Constraint Programming, Data Structures and Algorithms, Databases

Skills

- Languages:** Python, Java, JavaScript, TypeScript, C++, SQL
Web: React, Next.js, Node.js, Flask
Machine Learning: PyTorch, scikit-learn, Hugging Face Transformers, MONAI
Cloud & DevOps: AWS (Certified Cloud Practitioner), Docker, Git, GitHub Actions, Linux
Databases & Testing: PostgreSQL, Redis, Jest, Pytest, Locust

Experience

- PricewaterhouseCoopers (PwC)** | *Software Engineering Intern* Jun 2024 – Aug 2024
- Owned the data pipeline and evaluation harness for a **PyTorch BERT** sentiment classifier processing **10,000 daily customer reviews** for a retail client, built alongside two ML engineers; reached **F1 0.91** on the held-out test split
 - Deployed as a containerised **Flask REST API** on **AWS EC2** with gunicorn workers behind an Application Load Balancer; sustained **150 req/s** with **p95 latency under 500ms** in Locust load tests
 - Built **CloudWatch** dashboards on prediction drift and request volume, defining alert thresholds tied to weekly F1 regression as rollback triggers
 - Optimised preprocessing with **pandas** vectorisation, cutting daily batch processing time by **60%**
- PricewaterhouseCoopers (PwC)** | *Software Engineering Intern* Aug 2023 – Sep 2023
- Shipped an internal **client onboarding dashboard** in **React** and **Node.js** replacing a manual spreadsheet workflow for a financial services team; pilot timings showed onboarding falling from **3 days to 4 hours** across 200+ onboarded users
 - Designed a **PostgreSQL** schema with **role-based access control** across 4 user roles, working with a data scientist on requirements; integrated with PwC's internal SSO via **OAuth 2.0**
 - Wrote **CI/CD pipelines** in **GitHub Actions** with linting, automated test gates, and staged deployments to **AWS**; no failed releases across the deployment window post-adoption
 - Achieved **90%+ code coverage** across **45 unit and integration tests** (**Jest**, **Supertest**) spanning the REST API and data access layers

Projects

- Real-Time Collaborative Code Editor**
- Built a multi-user code editor in **React** and **Node.js** using **WebSockets** and **Operational Transform** for conflict-free concurrent editing, with syntax highlighting for 12 languages via CodeMirror
 - Implemented a **Redis** pub/sub layer for horizontal scaling across server instances, sustaining **50+ concurrent editors** at **p95 synchronisation latency under 100ms** on k6 benchmarks, with automatic reconnection on network loss
 - Added **JWT authentication**, per-room permission levels (owner/editor/viewer), and session persistence in **PostgreSQL** enabling users to rejoin documents with full edit history intact
- Database Query Optimiser**
- Built a **PostgreSQL** analysis tool in **Python** parsing query ASTs and **EXPLAIN plan** output to surface missing indexes; Dockerised for reproducibility
 - Across **200+ generated queries** over a 5M-row, 12-table schema, identified **21 missing indexes** cutting average query time by **65%** (worst case 2.4s to 420ms) and removing sequential scans on hot paths
- Fair Uncertainty Quantification for Selective Prediction in Medical Image Segmentation (MSc Dissertation)**
- Built a selective prediction framework deferring unreliable outputs to clinical review rather than failing silently, focused on calibration and distributional fairness of segmentation uncertainty
 - Traced the coverage-Dice trade-off on 9-class pelvic MRI segmentation with **nnU-Net** and **Monte Carlo Dropout**, reaching **0.89 Dice at 85% coverage**; compared entropy and mutual information as uncertainty aggregation strategies

Leadership

- St Andrews AI Society** | *Machine Learning Officer* Sep 2024 – Apr 2025
- Led weekly machine learning workshops for **50+ members** spanning technical and non-technical backgrounds, designing sessions tractable across a wide range of mathematical and programming experience